

CLAUDE MASTER BUILD PROMPT

QUALITY360 NEXT – PHASE 1

Act as a Principal Product Engineer, UX Architect, Enterprise Software Architect and Frontend Lead.

Build a production-quality web application called Quality360.

This is NOT a prototype.

This is NOT a wireframe.

This is NOT a mockup.

The application must look and behave like a real enterprise product that could be deployed to customers.

Use:

Frontend:

- Next.js
- TypeScript
- TailwindCSS
- shadcn/ui

Backend:

- Python FastAPI
- PostgreSQL

Architecture Requirements:

- Reusable Workbench Framework
- Workflow Engine
- Approval Framework
- Audit Trail Framework
- Notification Framework
- Instrument Integration Framework
- Quality Insights Framework

The first implemented business module is:

Incoming Material Inspection

Business Story:

A raw material lot arrives.

A sample is collected.

Tests are assigned.

Results are captured through:

- Instrument Import
- Manual Entry
- File Upload

Results are validated.

Quality Insights are generated.

The material is approved, held or rejected.

Create the following pages:

1. Quality Operations Dashboard
2. Incoming Material Inspection Queue
3. Incoming Material Inspection Workbench
4. Instrument Integrations
5. Master Data

Requirements:

- Modern enterprise UX
- Responsive design
- Glassmorphism-inspired cards
- Activity feed
- Timeline progress
- Real-time notifications
- Role-based actions
- Complete CRUD operations
- Audit trail

Instrument Simulation:

Provide realistic instrument interactions.

When importing results:

Show processing states:

Connecting... Verifying... Reading... Parsing... Validating... Completed...

Duration approximately 5 seconds.

Then populate realistic results.

Do not instantly display data.

Users must feel connected to a real instrument.

Quality Insights Panel:

Always visible.

Display:

Recommended Action Risk Level Supplier Health Historical Comparison Observations

Do not use the term AI.

Use business language.

Notification Requirements:

Every action must generate feedback.

Examples:

Receipt Created Sample Created Results Imported Validation Completed Approval Completed

Use professional toast messages.

Data Model:

Design production-ready entities:

Supplier Material Receipt Sample Test Result Approval Workflow Instrument Notification AuditLog

Important:

Do not hardcode Incoming Material Inspection logic.

Build a generic framework so future modules:

- Heat Chemistry
- Casting Quality
- Mechanical Testing

- MTC & Dispatch

can reuse the same architecture and components.

The resulting application should feel like a modern manufacturing quality platform and not a traditional LIMS.